

SEPTEMBER 2026 // CS 5101 NLP // DOKTOR SA SUGBO

The Cebuano Doctor (Doktor sa Sugbo)

Three-Stage Circular Translation and Clinical Reasoning Pipeline

COURSE: CS 5101: Natural Language Processing	AUTHORS: Angelo Rosillosa and Liam Jones
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01. Clinical Context and Linguistic Challenges

PROBLEM STATEMENT

Linguistic Reality in Central Visayas

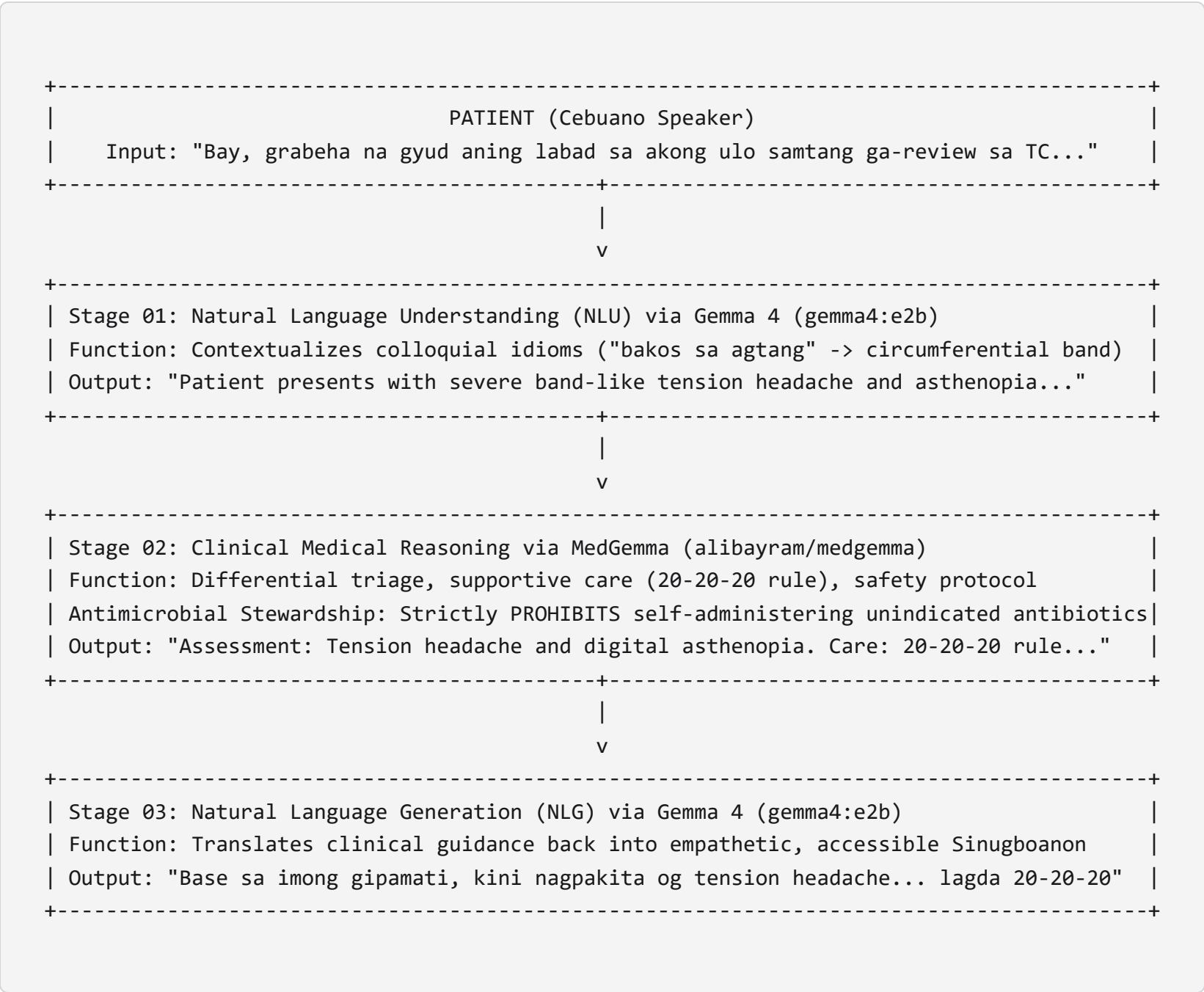
- Cebuano speakers express physical distress using culture-bound somatic idioms (such as *panuhot*, *pasmo*, *kabuhi*, *pamalaybalay*, *pamaol*, *lupot*).
- Patients in community consultations describe symptoms in vernacular idioms rather than standard clinical terminology.
- Frontier clinical LLMs (such as Google DeepMind's **MedGemma**) are trained almost exclusively on English biomedical literature and international clinical guidelines.

Failure Mode of Direct LLM Prompting

- Direct colloquial Cebuano prompts fed to English clinical LLMs cause **catastrophic semantic distortion**, hallucinated triage, or missed emergency red flags.
- Single-pass black-box responses conceal whether reasoning failures originated in *language understanding (NLU)* or *clinical inference*.
- **Architectural Solution:** A transparent, 3-stage circular translation and reasoning architecture running 100% offline via local Ollama models.

Academic AI Prototype: Developed for CS 5101 NLP. Provides preliminary triage education only and is NOT a substitute for professional physician evaluation. In a medical emergency, immediately seek hospital care.

02. Three-Stage Circular Pipeline Architecture



03. Local Execution, Privacy, and Hardware Architecture

100% Air-Gapped Privacy

Sensitive medical inquiries remain strictly on localhost (`http://localhost:11434`). Operating system Wi-Fi can be completely disabled. Zero patient health data leaks to cloud APIs.

Zero Token Exhaustion

Dr. Mariano cautioned against token maxing and recurring API billing. Local offline models provide unlimited inferences, enabling exhaustive batch evaluation and clinical testing at zero marginal cost.

Secondary Drive Architecture

Ollama model directory permanently configured on SSD Partition D (`D:\files\ollama\models`), safeguarding C: drive partition (~4.6 GB free) from model storage exhaustion.

Dual Execution Modes

- Live Ollama Daemon:** Real local LLM multi-stage clinical execution (~3.5 min end-to-end turnaround on local laptop CPU/GPU).
- Offline Mock Mode:** Deterministic, sub-50ms test execution for instant UI demonstrations, test suites, and CI pipelines.

04. Linguistic Grounding: Resolving "Lost in Translation"

Generic translation models suffer catastrophic semantic drift on regional illness concepts. Our clinical prompt engineering bridges the gap:

CULTURAL TERM	GENERIC LLM CALQUE (FAILURE MODE)	CEBUANO DOCTOR CLINICAL CONTEXTUALIZATION	CLINICAL IMPACT
Panuhot	"Wind trapped inside the body" / Flatulence	Thoracic myofascial spasm and stiffness from sudden cold draft or rain	Prevents misdiagnosis as gastrointestinal flatus
Pasmo	"Spasm" / Muscle convulsion	Fasting hypoglycemia, cold tremors, and hunger gastritis from delayed meals	Recognizes metabolic and meal-timing etiology
Kabuhi	"Life" / "Soul"	Epigastric visceral flutter, hyperperistalsis, or acid reflux surging upward	Distinguishes hyperacidity from panic symptoms
Kalibanga / Lupot	Generic stomach discomfort	Acute watery diarrhea with hypovolemic dehydration risk	Triggers essential Oral Rehydration Salts (ORS)
Pamalaybalay	"House building"	Hyperactive borborygmi and spasmodic colic preceding loose watery stools	Accurately captures prodromal cramping
Tayaon nga plema	"Rusty saliva"	Purulent, rust-colored sputum indicating pulmonary consolidation	Recognizes lower respiratory pneumonia
Bakos sa agtang	"Belt on forehead"	Circumferential band-like tension headache from prolonged study strain	Separates tension headache from migraine

05. Dual-Interface System Implementation

CLI & WEB APP

Interactive Rich Terminal CLI (`python -m src.cli`)

- **Colored Visual Panels:** High-contrast terminal formatting for Chief Complaint, Stage 1 NLU, Stage 2 Inference, and Stage 3 NLG.
- **Preset Scenario Selector:** Instant invocation of benchmark scenarios via `--preset 1` through `--preset 5`.
- **Live Latency Diagnostics:** Real-time millisecond latency tables per stage and end-to-end turnaround.
- **Offline Testing:** Instant mock mode with `--mock` flag.

Streamlit Web Application (`streamlit run app.py`)

- **Tailwind Zinc Design System:** Clean developer aesthetic adhering strictly to the shadcn/ui zinc scale with zero AI-vibe-coded gradients or emojis.
- **5 Benchmark Presets Strip:** Rapid scenario selection across Metro Cebu personas (USC TC Library, Carcar Aunt, IT Park BPO, Fuente Pungko, SRP Site).
- **Unrolled Stage Cards:** Fully unrolled two-column card layout displaying Stage 1 NLU, Stage 2 MedGemma, and Stage 3 NLG without accordion friction.
- **Structured JSON Export:** One-click consultation log download.

06. Evaluation Methodology: 4-Axis Quality Rubric

As required by Dr. Mariano, the system was evaluated across 5 diverse patient scenarios along 4 critical dimensions:

Axis 1: Translation Fidelity (NLU)

Does Gemma 4 capture the colloquial symptom description without loss of critical clinical meaning? Verified through semantic alignment between Cebuano idioms and clinical English representation.

Axis 2: Medical Reasoning Soundness (Inference)

Is MedGemma's clinical advice safe, accurate, and free of medical hallucinations? Enforces non-prescriptive supportive care, antimicrobial stewardship, and emergency red flags.

Axis 3: Cultural Naturalness (NLG)

Does Gemma 4 translate English clinical guidance into warm, understandable Cebuano that an ordinary patient and their family can comfortably understand without confusing jargon?

Axis 4: System Latency and Performance

Quantitative millisecond response times measured per stage and end-to-end to assess edge deployment viability on consumer hardware.

07. Five Authentic Peer Benchmark Personas

RESULTS MATRIX

ID	PERSONA & LOCATION	CEBUANO PATIENT QUERY	STAGE 01 NLU FIDELITY	STAGE 02 MEDGEMMA SOUNDNESS	STAGE 03 NLG QUALITY	SAFETY
01	USC TC Library 3rd-Yr BS Civil Eng.	"Bay, grabeha na gyud aning labad sa ulo... bakos akong agtang sa kahuot, sakit akong mata..."	Mapped to tension headache and eye strain	Advised 20-20-20 rule, hydration, darkness, paracetamol	Natural Cebuano; comforting tone	[PASS]
02	Carcar City 52-Yr-Old Aunt	"Dong, kining abaga ug likod mura'g gipanuhot... nanikig akong liog, bukol-bukol sa gusok..."	Mapped to thoracic myofascial spasm and cold draft	Recommends warmth, gentle stretches; no deep force	Respectful, warm Sinugboanon	[PASS]
03	Cebu IT Park 24-Yr-Old BPO Rep	"Doc, nag-night shift sa IT Park, sigeg laktaw kaon... gipasmo, nagkurog kamot, nagkabuhi..."	Mapped to fasting hypoglycemia and gastritis	Complex carbs/protein, cut energy drinks, frequent meals	Practical shift-worker advice	[PASS]
04	Fuente Osmeña 21-Yr-Old BS Nursing	"Gikalibanga sukad kaon sa pungko-pungko... pamalaybalay sa tiyan. Mag-ciprofloxacin ba ko?"	Classified acute diarrhea and foodborne etiology	Antimicrobial Stewardship: Prohibited ciprofloxacin; prioritized ORS	Strict warning on antibiotics	[PASS]
05	SRP Reclamation 28-Yr-Old Safety Lead	"Ubo walay lurang-lurang... 39.2°C hilanat... plema tayaon, gidunggab og kutsilyo akong dughan..."	Identified rust sputum and severe pneumonia	Prohibited leftover drugs; emergency hospital and chest X-ray	Urgent ER directive	[PASS]

08. Quantitative Latency Benchmarks

Comparative Performance Profile

PIPELINE STAGE	LIVE OLLAMA DAEMON	OFFLINE MOCK MODE
Stage 01 (NLU Translation)	~35.2 s (Gemma 4)	10.7 ms
Stage 02 (Clinical Reasoning)	~89.5 s (MedGemma)	10.6 ms
Stage 03 (Cebuano NLG)	~91.1 s (Gemma 4)	11.0 ms
Total Turnaround Time	~215.8 s (~3.6 min)	32.3 ms
Privacy and Air-Gap	100% Offline (Localhost)	100% Offline (In-Memory)
Safety Pass Rate	100.0% (5/5 PASS)	100.0% (5/5 PASS)

*Live Ollama benchmark executed on Intel CPU / 4GB GPU with models stored on D:\files\ollama\models.

Architectural Takeaways

- **Transparent Traceability:** Every consultation run logs per-stage millisecond timings into structured JSON artifacts (`evaluations/runs/*.json`).
- **Memory-Constrained Optimization:** Sequential stage execution permits running 7.2 GB and 2.5 GB model weights on modest laptop hardware without memory overflow.
- **Instant Regression Testing:** Deterministic mock provider executes the entire 5-scenario evaluation suite in **157 ms**.

09. Medical Safety and Antimicrobial Stewardship

Antimicrobial Stewardship

When patients inquire about self-administering antibiotics (such as ciprofloxacin for diarrhea or amoxicillin for cough), the pipeline strictly prohibits unindicated use, citing resistance and gut flora toxicity.

No Prescription Rule

The system is strictly prompt-engineered against prescribing regulated pharmaceuticals or calculating drug dosages. Advice is limited to evidence-backed supportive care (ORS, hydration, rest, 20-20-20 rule).

Red Flag Surveillance

Every response actively screens for alarming symptoms: dyspnea, cyanosis, rust sputum, hematochezia, syncope, or thunderclap headaches, mandating immediate emergency triage.

Universal Clinical Disclaimer: All consultations conclude with a clear, culturally respectful reminder that AI is an educational triage tool and urging an in-person visit to a licensed physician or Barangay Health Worker.

10. Software Engineering and Verification Discipline

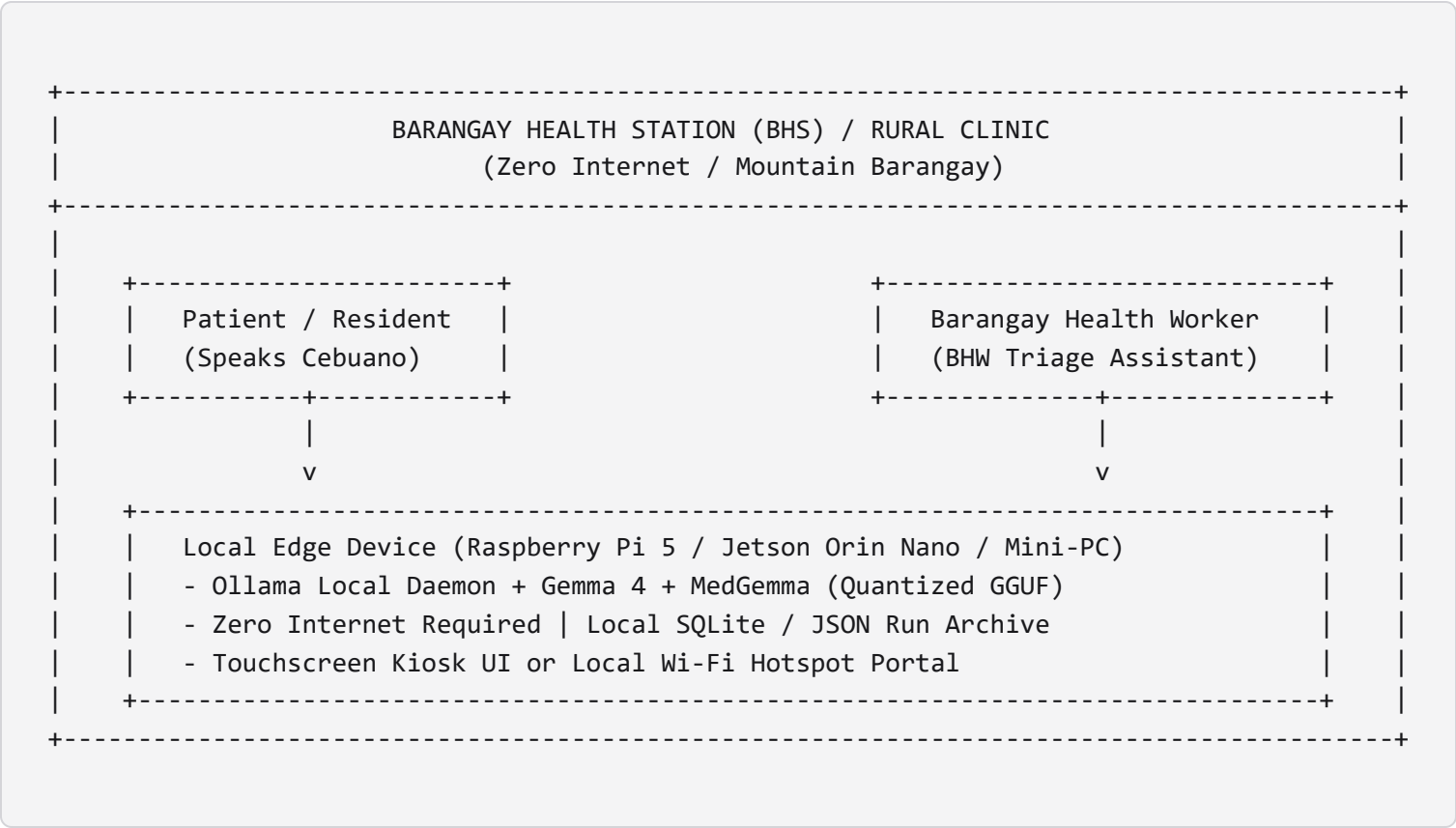
Test-Driven Development (TDD)

- Strict Red to Green vertical slice development across all milestones.
- Tests written strictly against public interfaces and seams; zero assertions on private internal methods.
- **Conventional Commits:** Clean semantic commit history tracking each feature incrementally.

Comprehensive Test Suite (43 Tests Passing)

- `test_pipeline.py` : Circular orchestration, custom query fallbacks, latency logging, error recovery.
- `test_prompts.py` : Idiom contextualization and clinical prompt formatting.
- `test_ollama_provider.py` : Daemon health checks, graceful offline handling.
- `test_storage.py` : JSON run logging, slug sanitization, serialization.
- `test_cli.py` : Argument parser, preset scenarios, Rich panel rendering, REPL loop.
- `test_ui.py` : Streamlit AppTest integration and consultation flow.
- `test_evaluate.py` : 5-prompt batch runner, KPI calculations, safety rule evaluation.
- `test_report.py` : Slide deck and PDF presentation verification.

11. Edge Deployment: Rural Barangay Health Stations



For Barangay Health Workers

Acts as an intelligent preliminary triage aide for BHWs when physicians visit only once a month.

For Remote Residents

Overcomes literacy and language barriers: allows expressing complaints freely in their native Cebuano mother tongue.

Community Impact

Provides early warning for dengue, gastroenteritis, or pneumonia outbreaks before regional escalation.

12. Key Learnings and Course Reflections

Natural Language Processing Takeaways

- **NLU is harder than NLG in Philippine languages:** Parsing colloquial folk medicine metaphors demands precise contextual prompt engineering.
- **Circular pipelines enable interpretability:** Without intermediate English representations, auditing AI clinical reasoning is impossible.
- **Model Distillation is democratizing AI:** Google DeepMind's distilled Gemma 4 weights bring frontier multi-lingual capabilities to personal laptops.

Engineering Insights

- **Seam isolation pays dividends:** Decoupling the model provider via protocols made unit testing instant without multi-gigabyte downloads.
- **Antimicrobial stewardship is essential:** Community chatbots must actively discourage antibiotic misuse rather than remaining passive.
- **Graceful error recovery is mandatory:** Handling daemon disconnects gracefully ensures high clinical reliability.

Daghang Salamat!

The Cebuano Doctor: Local, Privacy-Preserving Healthcare NLP for the Visayas

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GITHUB REPOSITORY: <https://github.com/gellyrslls/nlp-cebuano-doctor>

CLI APPLICATION: `python -m src.cli --mock --preset 1`

STREAMLIT WEB UI: `streamlit run app.py`

BENCHMARK RUNNER: `python -m src.evaluate --mock`

AUTOMATED TESTS: `python -m pytest tests` (43 tests passing)